TURKIC INTERNATIONAL MATHEMATICAL OLYMPIAD

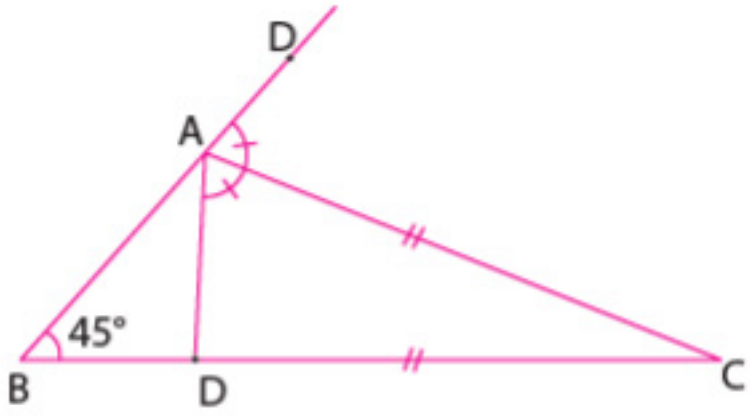
TIMO

GRADE 9

Mathematical Horizons: Uniting Turkic Minds, Expanding Global Genius.

SCHOLARSHIP FROM TURKISH UNIVERSITIES

1.

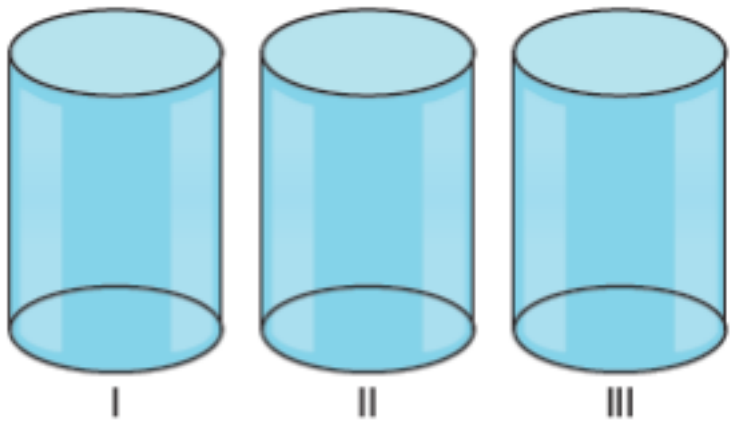


In the igure above, $m(\widehat{DBC}) = 45°$

$|AC| = |DC|$ and $m(\widehat{DAC}) = m(\widehat{ADC})$.
According to this what is $m(ACD)$?

A) 42 B) 40 C) 37,5 D) 32,5 E) 30

2.



Three equal containers of 12 litres each are completely filled with water. Ali fills the water in container I with enough amount o $\sqrt{2}$ litre bottles, Berk fills the water in container II with enough amount o $\sqrt{3}$ litre bottles and Can fills the water in container III with enough amount o $\sqrt{5}$ litre bottles so that as many bottles as possible are ull. The amounts o water remaining in containers I, II and III are a, b, and c litres respectively.
According to this which of the following gives the correct order of a, b and c?

A) $a<b<c$ B) $a<c<b$ C) $c<b<a$

D) $c<a<b$ E) $b<a<c$

3.

I II

Four marbles with the numbers 5, 8, 12 and 15 written on them are put in box I and five marbles with the numbers 4, 9, 18, 20 and 25 written on them are put in box II. The marbles are identical except or the numbers on them. Ali first draws 1 marble from box I and then draws 1 marble from box II.

What is the probability that the sum of the numbers on these two marbles is an odd number?

A) $\frac{4}{5}$ B) $\frac{3}{5}$ C) $\frac{1}{2}$ D) $\frac{7}{10}$ E) $\frac{3}{4}$

4. A 2024 digits natural number which has all its digit 2 is given.
What is the sum of the digits of the number obtained by multiplying this number by 2222?

A) 14163 B) 14170 C) 14177 D) 14184 E) 14191

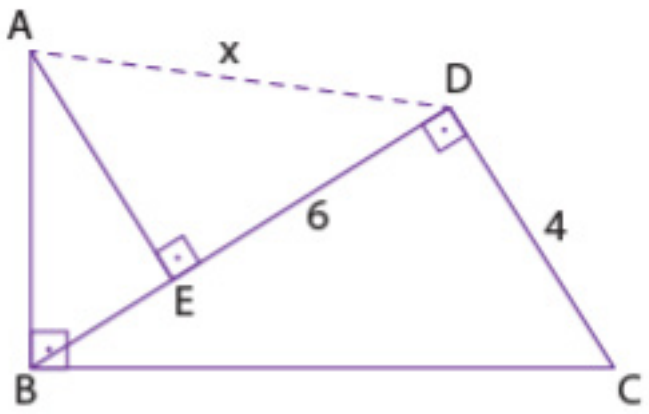
5. Given the inequality

$$|2a - 24| + |b - a| \leq 2$$

How many different ordered pairs (a,b) are there in integers?

A) 1 B) 3 C) 5 D) 7 E) 8

6.



In the figure above

$2|AB| = |BC|$

$|ED| = 6$ cm, $|DC| = 4$ cm

$m(\widehat{ABC}) = m(\widehat{AED}) = m(\widehat{BDC}) = 90°$

A) $2\sqrt{6}$ B) $4\sqrt{2}$ C) $4\sqrt{3}$ D) $2\sqrt{13}$ E) 8

7. Efe, who goes for a ride with his bicycle in the park, is at the entrance of the park at 08.00 am. There is one of each kind of tree in this park. Efe passes by acacia tree at 08.20 am, plane tree at 08.35 am, poplar tree at 09.05 am, cedar tree at 09.45 am, hornbeam tree at 10.45 am and reaches the exit of the park at 11.00 am. After resting for half an hour, it sets off towards the entrance from the same road it came from, at a constant speed of half the speed of its arrival. According to this, at what time does he pass by the plane tree on the way back?

A) 16.40 B) 16.20 C) 16.00 D) 15.50 E) 15.40

8. $A \subseteq B$ Given the set A is a subset of the set B. If the sum of the number of the subsets of the sets A and B is 68, How many subsets of the set B do not contain the set A?

A) 16 B) 24 C) 32 D) 36 E) 48

9. For the natural numbers a and b, if a • b=24 then what is the smallest value of 2a+3b?

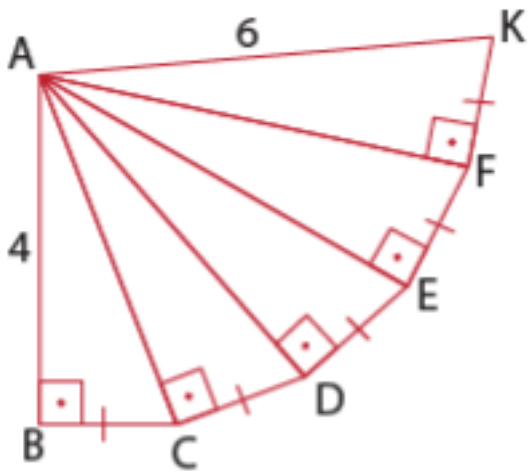
A) 23 B) 24 • C) 25 D) 26 E) 30

10. If $a+\frac{1}{b}=x, b+\frac{1}{c}=y$ and $c+\frac{1}{a}=z$ then find the expression $abc+\frac{1}{abc}$ in terms of x, y and z?

A) $xyz+x+y+z$ B) xyz C) $x+y+z$

D) $xyz-x-y-z$ E) $x+y+z-xyz$

11.



Above, one of the right side of the triangles are equal to each other as shown in the figure.
|BC|=|CD|=|DE|=|EF|=|KF|,
|AB|=4 cm and |AK|=6 cm
According to this what is the length |AE| in cm?

A) $2\sqrt{5}$ B) $3\sqrt{2}$ C) $\frac{9}{2}$ D) $2\sqrt{7}$ E) $4\sqrt{2}$

12. In the table, some of whose numbers are deleted, the numbers are written in such a way that the sum of the three numbers written vertically and horizontally is equal to each other.

4				3
7		5		
6				

What is the sum of the deletede numbers in the table?

A) 43 B) 39 C) 37 D) 31 E) 28

13. When at most how many digits of the 2024-digit number 20242024 ... 2024 are deleted, the sum of the digits of the remaining number is 2024?

A) 1430 B) 1490 C) 1496 D) 1518 E) 1524

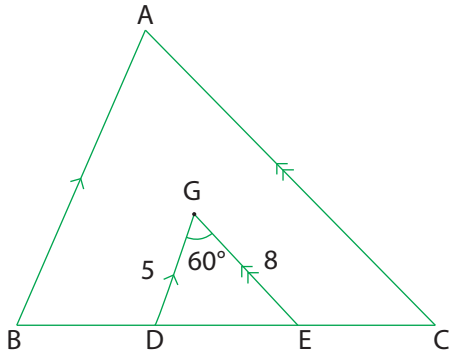
14. What is the product of the digits of the value of x satisfying the equation $x^x=3^{x+54}$?

A) 9 B) 12 C) 14 D) 15 E) 0

15. Veysel and Şengül is making a study programme together. Each day, Veysel is solving 3/2 of the number of questions he solved in the previous day and Şengül is solving 1/4 of the number of questions she solved respectively in the previous day.
Since they solved an equal number of questions on the 6th day of the programme, on the first day, how many times the number of questions solved by Şengül is the number of questions solved by Veysel?

A) 3^5 B) $2 \cdot 3^5$ C) $4 \cdot 3^5$ D) 6^5 E) 12^5

16.



The point G is the center of the gravity of the triangle ABC.
If GD//AB, GE//AC, $m(\widehat{DGE})=60^o$, |GD|=5 cm and |GE|=8 cm then what is the perimeter of the triangle ABC in cm?

A) 36 B) 48 C) 54 D) 60 E) 69

17. A battery is fully charged after 45 minutes when charged with machine A, 40 minutes when charged with machine B and 30 minutes when charged with machine C.
When this battery is charged first with A for a min, then with B for b min and then with C for c min, 30% of the battery is full, how many different values does the sum of the positive integers a, b and c take?

A) 1 B) 2 C) 3 D) 4 E) 5

18. A product is sold at 20 ₺ when packed in 3 kg A packs and at 32 ₺ when packed in 5 kg B packs. If a person who has 485 ₺, buys 74 kg of these packages in total, at most how many packages of product can he buy?

A) 24 B) 22 C) 20 D) 18 E) 16

19. For the system / Для равенств

$xyz = 1$, $x + \frac{1}{y} = 5$, $y + \frac{1}{z} = 21$ and / и $z + \frac{1}{x} = \frac{a}{b}$

A) 33 B) 34 C) 35 D) 36 E) 38

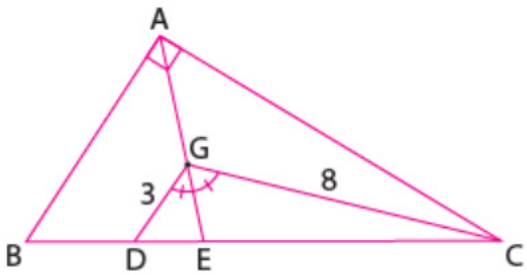
20. For all reel numbers $a_1, a_2, a_3 \ldots, a_n$ and $b_1, b_2, b_3 \ldots, b_n$ the inequality.

$(a_1 \cdot b_1 + a_2 \cdot b_2 + \ldots + a_n \cdot b_n)^2 \leq (a_1^2 + a_2^2 + \ldots + a_n^2)(b_1^2 + b_2^2 + \ldots + b_n^2)$ holds.

According to this what is the greatest value of the reel number $\sqrt{x-1} + \sqrt{2x-5} + \sqrt{15-3x}$?

A) 5 B) $\sqrt{26}$ C) $3\sqrt{3}$ D) $2\sqrt{7}$ E) $\sqrt{29}$

21.



In the figure BAC is a right triangle with m(BAC)=90º and the point G is its' center of the gravity. If |DG|=3 , |GC|=8 , m(DGE)=m(EGC) and the points A, G and E are collinear find the ratio $\frac{|BD|}{|AG|}$?

A) $\frac{7}{8}$ B) $\frac{9}{10}$ C) $\frac{11}{12}$ D) $\frac{14}{15}$ E) $\frac{15}{16}$

22. For the reel numbers x and y if $\left(2x+\sqrt{1+4x^2}\right)\left(3y+\sqrt{1+9y^2}\right)=1$ then find the value of $(2x+3y)^2$?

A) 0 B) 1 C) 2 D) 3 E) 4

23. Given pozitive reel numbers x, y and z if

$$x+y+z=20$$
$$\sqrt{256-x^2}+\sqrt{81-y^2}+\sqrt{16-z^2}=21$$

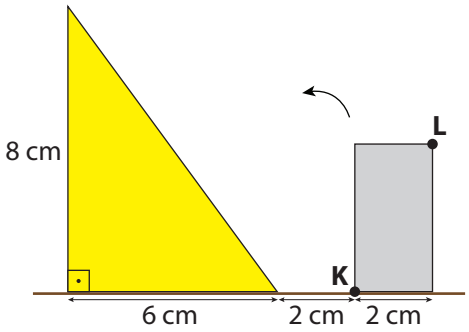
Then find the value of $\frac{x+y}{z}$?

A) 6,25 B) 6,5 C) 6,75 D) 7 E) 7,5

24. How many different positive integer pairs (x,y) are there satisfying the equation $x^3=3^y-1$?

A) 0 B) 1 C) 2 D) 3 E) 4

25.



The right side lengths of the yellow right triangle in the figure are 6 cm and 8 cm. The short side of the grey rectangle drawn on the same ground as the triangle is 2 cm. When the rectangle is rotated 90° around point K in the direction of the arrow, point L of the rectangle lies on the hypotenuse of the triangle.
Since the distance between the triangle and the rectangle is 2 cm, what is the area of the region where the triangle and the rectangle intersect after the rectangle is rotated in cm^2?

A) $\frac{3}{2}$ B) $\frac{3}{4}$ C) $\frac{5}{2}$ D) 2 E) $\frac{7}{4}$